
Identification of Replicable Passive Sensing Biomarkers for Depression in Longitudinal Multi-Wave Cohorts

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Abstract

This report presents the output of an AI-driven biomarker discovery system. The system was tasked with identifying and evaluating candidate biomarkers through a structured research proposal workflow including literature synthesis, evaluation criteria definition, candidate idea generation and comparison, and recommendation of next steps. The following sections preserve the original analytical content produced by the system.

1 Research Goal

The primary research goal is to identify and validate replicable, passive mobile sensing biomarkers for depression within the high-dimensional GLOBEM dataset. This task involves extracting meaningful digital signatures from 5,508 features generated by the RAPIDS pipeline across seven distinct modalities: location, screen state, sleep architecture, Bluetooth proximity, telephony, physical activity (steps), and WiFi connectivity. A central challenge of this research is ensuring the generalizability of these biomarkers across longitudinal "waves." The dataset encompasses four distinct waves (N=704 participants, 8,225 observations), bridging the critical pre-pandemic (Waves 1-2) and post-pandemic (Waves 3-4) eras.¹ To achieve this goal, the research requires: Sparsity Management: Handling features with high missingness rates while adhering to a strict 70% exclusion threshold. Leakage Prevention: Implementing rigorous cross-validation grouped by User ID and excluding all auxiliary target labels to ensure the model captures genuine behavioral traits rather than identity "fingerprints" or circular logic. Statistical Rigor: Utilizing univariate analysis (Mann-Whitney U with Benjamini-Hochberg FDR correction) and multivariate modeling (LightGBM, XGBoost, Random Forest, and Logistic Regression).

Validation: Grounding results in a 11-test validation battery for binary outcomes, requiring biomarkers to demonstrate replication in at least two waves and maintain predictive stability across the temporal boundary of the COVID-19 pandemic.

2 Evaluation Criteria

The identification of depression biomarkers from mobile sensing data requires a multi-faceted evaluation framework to ensure that the results are not only statistically significant but also clinically actionable and technically robust. The following criteria serve as the benchmark for evaluating proposed methodologies:

2.1 Cross-Wave Replicability and Generalizability

Proposed ideas must demonstrate Longitudinal Consistency, where biomarkers maintain predictive performance across at least two distinct study waves. A critical metric is Temporal Generalization: the ability of a model trained on pre-pandemic data (Waves 1+2) to maintain accuracy when applied to the altered behavioral landscapes of the pandemic and post-pandemic eras (Waves 3+4). This

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necessitates Directional Stability, ensuring the association between a feature and depression does not flip signs across environments, and Scale Invariance, prioritizing relational metrics (ranks, ratios) over absolute volumes (total steps) that are sensitive to lockdowns.²

2.2 Methodological Rigor and Leakage Prevention

Integrity in the machine learning pipeline is paramount. Proposals must strictly exclude all non-primary target_* columns to prevent Label Leakage. Evaluation must utilize Subject Isolation (e.g., GroupKFold) to ensure the model does not "memorize" specific individuals. Furthermore, all feature selection and normalization must prevent Temporal Leakage by using expanding windows or ensuring that no future test-set statistics inform the training phase.

2.3 Statistical Validation and Sparsity Management

Given the high-dimensional nature of the RAPIDS features (5,508 features), proposals must adhere to the 70% Missingness Rule, excluding sparse features while interpreting Informative Missingness (MNAR). Methodologies must apply Multiple Testing Corrections (Benjamini-Hochberg FDR) per modality to mitigate Type-I errors. Mathematical validity is also essential; ideas must avoid undefined operations on categorical data or spectral aliasing when attempting to extract 24-hour rhythms from aggregated data.

2.4 Technical and Hardware Alignment

Models must account for Hardware Heterogeneity, distinguishing between behavioral signals and technical noise created by differing

background execution policies on iOS versus Android. Proposed features must align with the Data Granularity available (e.g., 6-hour epochs rather than minute-level event logs) and use only the seven authorized RAPIDS modality groups.

2.5 Behavioral Interpretation and Clinical Utility

Biomarkers must be grounded in recognized psychological frameworks, such as the Research Domain Criteria (RDoC).³ For instance, mapping location variance to anhedonia or morning screen latency to psychomotor retardation.^{4, 5} The proposed systems should demonstrate Clinical Accuracy and provide added value over simple volume-based baselines through the 11-test battery for binary outcomes.

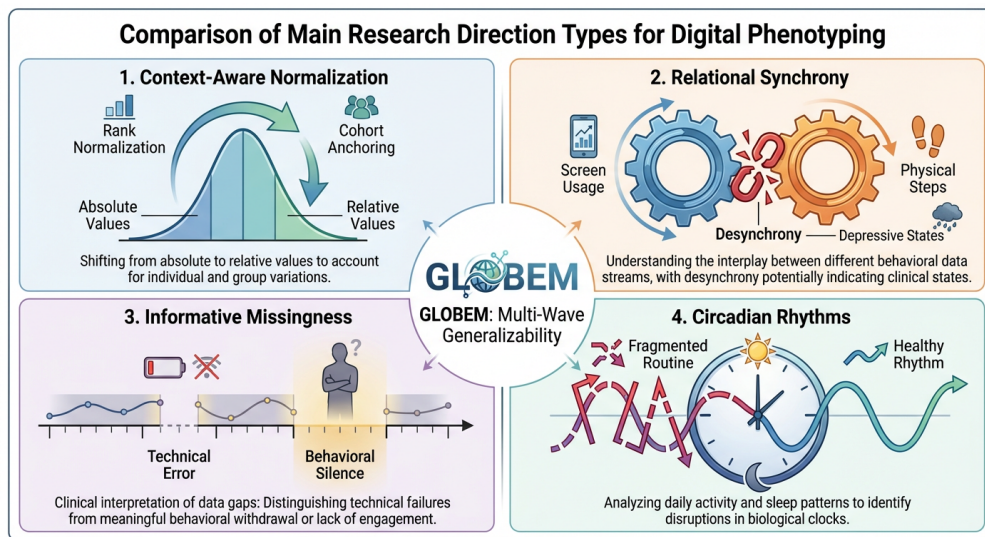


Figure 1: Conceptual diagram 1 from the biomarker discovery analysis.

3 Main Research Directions

The following research directions represent the state-of-the-art strategies for extracting stable digital phenotypes from the GLOBEM/RAPIDS infrastructure.

3.1 Context-Aware Normalization and Invariant

Engineering

To overcome the "domain shift" caused by macro-environmental shocks like COVID-19, research is shifting toward Relational Metrics. Instead of absolute step counts, which plummeted for all participants during lockdowns, research focuses on the ratio of digital engagement to physical activity (Screen-to-Step ratio) or Wave-Cohort Z-Scoring. These methods anchor an individual's behavior against their peers' current distribution, ensuring that a campus-wide mobility drop is not misidentified as widespread clinical depression.

3.2 Relational Biomarkers and Multi-Modal Synchrony

Modern research explores the "coupling" between different behavioral streams. A primary direction is Digital-Physical Desynchrony, which measures the lag between "digital awakening" (screen unlock) and "physical movement." Other approaches model modalities as nodes in a Topological Behavioral Network, where a loss of connectivity between social, spatial, and physical nodes serves as a robust marker for the systemic breakdown of daily routines.

3.3 Informative Missingness and "Digital Silence"

Rather than viewing data gaps as a failure of collection, research now treats missingness as a Clinical Signal. By distinguishing between technical failures (dead battery) and Behavioral Silence (device is active but the user is not interacting), researchers can identify "Digital Blackout" signatures as proxies for psychomotor retardation and social withdrawal.

3.4 Circadian Integrity and Rhythmic Stability

Depression is fundamentally linked to the breakdown of biological rhythms.⁶ Key directions include Off-Schedule Circadian Power (OSCP), which isolates behavior during "unstructured time" (weekends/evenings) where intrinsic motivation is more visible than during rigid academic schedules. Researchers also utilize Morning Inertia Metrics to capture the "morning heaviness" and psychomotor slowing characteristic of melancholic phenotypes.⁷

Comparison of the four primary research frameworks for cross-wave depression biomarker discovery. Note: The AI image generation process (Beta) is experimental and may not description.

3.5 Rigorous Multi-Wave Validation

The final pillar of current research is the move toward Identity-Neutral Filtering and Stability Selection. By using Sparse Group Lasso and conducting "Identity Audits," researchers ensure that models do not overfit to individual "fingerprints" (like specific GPS coordinates) but instead capture generalizable behavioral motifs that survive the transition across independent longitudinal cohorts. This ensures that only the most robust biomarkers are reported as having passed the GLOBEM 11-test battery.

4 Candidate Ideas for Replicable Biomarker

Discovery The following sections provide detailed descriptions of the ten candidate research frameworks proposed for identifying replicable depression biomarkers within the GLOBEM dataset. Each proposal addresses the core challenges of high-dimensional feature selection, macroenvironmental shifts, and cross-wave generalizability.

4.1 Idea 1: Resilient Modality-Expert (RME) Framework

The Resilient Modality-Expert (RME) Framework is an architectural pipeline designed to isolate a "Resilient Behavioral Core" of biomarkers that maintain predictive power across the pre- and post-COVID eras. The primary innovation of RME is the Resilience Score (R -score), a heuristic calculated as the product of a feature's mean univariate AUC and its consistency (inverse coefficient of variation) across waves. This scoring system prioritizes features that are not only predictive but also stable during environmental shocks. To handle the pandemic-induced domain shift, RME utilizes Wave-Cohort Z-Scoring, which normalizes participants relative to their peers within the same wave, effectively treating the lockdown as a batch effect. The modeling phase employs a Hierarchical Late Fusion architecture. In this system, individual "Modality Experts" (e.g., a Location Expert, a Sleep Expert) are trained using a variety of algorithms, including LightGBM and XGBoost. The final depression probability is a weighted average of these experts. This approach allows for clinical robustness; if a specific sensor fails or suffers from extreme missingness (such as Bluetooth on iOS), the gating mechanism reduces that expert's weight, allowing the model to degrade gracefully rather than failing entirely. Motivation: RME is motivated by the observed 18.9% performance drop in static models applied across GLOBEM waves.⁸ By decoupling feature stability (via R -score) from final ensemble weights, it addresses the "identity fingerprinting" problem where models learn wave-specific artifacts rather than clinical signals. Category

Evaluation Description

Strengths

Robust handling of hardware failure via late fusion; strong mathematical defense against concept drift using the R -score.

Weaknesses

Some proposed features (e.g., Sleep Midpoint Drift) require circular statistics to avoid wrap-around errors; reliance on raw Bluetooth counts may be confounded by IoT density.

Missingness

Adheres to the $\geq 70\%$ exclusion rule; gates missing modalities dynamically.

Generalizability

High; specifically designed for W1+2 to W3+4 transition through wave-cohort normalization.

Judgement: RME provides a highly sophisticated architectural solution to longitudinal sensing, though its success depends on refining specific feature engineering to account for circularity and environmental noise.

The Resilient Modality-Expert (RME) Framework uses modality-specific experts and a stability-based resilience score to ensure cross-wave replication. Note: The AI image generation process (Beta) is experimental and may not be

4.2 Idea 2: Ratio Invariance Principle

The Ratio Invariance Principle posits that absolute behavioral volumes (e.g., total steps, total travel distance) are fundamentally confounded by macro-environmental forces that scale the entire population's baseline. To address this, the framework prioritizes biomarkers expressed as withinperson ratios, which are algebraically invariant to such scaling. For example, the Asymmetry Coefficient (AC) for communication directionality—calculated as the ratio of (outgoing - incoming) to total calls—remains stable even if the total call volume for the entire cohort drops due to external factors. This idea focuses on three mechanistically grounded groups: Communication Passivity (capturing the shift from agency to reactivity), Sleep Temporal Disorganization (using non-parametric metrics like Interdaily Stability), and Circadian Social Rhythm Collapse (analyzing the spectral power of Bluetooth interactions). By focusing on individual-level proportions rather than absolute magnitudes, the model aims to capture the individual biological signal of depression rather than the environmental context of the data collection wave. Motivation: Evidence from the GLOBEM benchmark shows that Random Forest models can identify which wave a participant belongs to with 91.1% accuracy based on absolute features.⁹ This suggests that models often learn the "temporal context" of the pan-

demic rather than the clinical state. Ratio invariance is proposed to "cancel out" this environmental scaling factor. Category Strengths

Evaluation Description

Theoretically grounded approach to removing environmental confounding; high interpretability for clinical phenotypes like anhedonia.

Category

Evaluation Description

Weaknesses

Assumption of linear behavioral scaling during lockdowns may be oversimplified; iOS privacy restrictions on call logs may lead to high missingness in the Communication modality.¹⁰

Missingness

Uses strict exclusion (>70%) and GBDT-based routing for remaining gaps.

Generalizability

Strong theoretical potential for zero-shot replication across environmental boundaries.

Judgement: This framework represents a significant conceptual improvement over raw volume-based features, though its viability is limited by the physical constraints of mobile OS data collection (e.g., iOS call log restrictions).

The Ratio Invariance Principle prioritizes within-person

4.3 Idea 3: Harmonized Behavioral Topology (HBT)

The Harmonized Behavioral Topology (HBT) framework focuses on the "structural integrity" of daily rhythms. Its primary features are the Relative Synchrony Index (RSI) and Circadian-Informed Missingness (CIM). RSI measures the statistical coupling (via Mutual Information) between digital engagement and physical mobility. A low RSI signifies "Functional Decoupling"—where a user engages in habitual screen use that is disconnected from their physical surroundings, serving as a proxy for ruminative withdrawal. CIM is a novel approach to Informative Missingness. It specifically identifies "Active Bed-Stagnation" by detecting periods where high-energy sensors (steps) are missing or zero while low-energy sensors (WiFi or screen) remain active. This transforms technical data gaps into a diagnostic signal for psychomotor retardation. The framework uses

Sparse Group Lasso for feature selection to ensure that only a core set of topological features are passed to the final LightGBM ensemble. Motivation: HBT addresses the 18.9% performance drop by moving away from absolute thresholds to relative topological metrics. It is specifically motivated by the need to treat missing data as a "Missing Not At Random" (MNAR) signal of clinical state.

Category

Evaluation Description

Strengths

Innovative use of missingness as a clinical proxy (CIM); high interpretability through RDoC mapping.

Weaknesses

Calculating Mutual Information on sparse 6-hour epoch data is mathematically prone to finite-sample bias;¹¹ RSI may be unstable without high-resolution timestamps.

Missingness

Unique approach: treats specific missingness patterns as a feature rather than noise.

Generalizability

Focused on environment-invariant "structural" features.

Judgement: HBT offers a powerful conceptual shift in handling missing data, but the statistical implementation of its synchrony metrics requires careful calibration to avoid noise in sparse longitudinal streams.

4.4 Idea 4: Cross-Modal Behavioral Network Topology

This idea treats the seven RAPIDS modalities as nodes in a Temporal Network. Instead of modeling 5,508 features independently, it uses Canonical Correlation Analysis (CCA) to define "behavioral coupling" as edge weights between modalities (e.g., the link between Sleep and Screen state). From this network, structural metrics are extracted, such as the Fiedler value (representing behavioral fragmentation) and node centrality. The hypothesis is that while absolute volumes (like steps) shift, the internal structural relationships—the "topology"—of a user's routine remain invariant. For instance, a depressed individual might show "Behavioral Decoupling," where digital activity becomes desynchronized from physical movement, regardless of whether they are in a prepandemic or lockdown environment. Motivation: This systems-level approach is motivated by evidence that interactive effects (e.g., high screen time combined with low physical movement) are stronger predictors of depression than individual features. Category Strengths

Evaluation Description

Captures complex inter-modality dependencies; bypasses macro-environmental confounds through structural invariants.

Category

Evaluation Description

Weaknesses

CCA requires $N \geq P$, which is difficult to achieve over short rolling windows; high-dimensional intra-modality covariance can lead to singular matrices.

Missingness

Network metrics are dynamically normalized by the number of active nodes.

Generalizability

Theoretically high, but technically difficult to implement without massive dimensionality reduction.

Judgement: A highly novel paradigm shift, but currently faces significant mathematical hurdles due to the sample-to-feature ratios required for stable network extraction.

4.5 Idea 5: Circadian-Relative Resilience (CRR)

Framework The CRR Framework focuses on a participant's relative behavioral standing within their cohort, specifically adjusted for hardware-induced noise. It introduces OS-Stratified PASK (Population-Anchored SocialKinesthetic) features, which rank mobility and social activity strictly within OS-matched sub-groups (iOS vs. Android). This prevents the model from misinterpreting the stricter background restrictions of iOS as clinical social withdrawal. Additionally, CRR uses Trait-Anchored Circadian Entrainment Entropy (TA-CEE), which measures deviations from a 14-day rolling personal baseline. This distinguishes clinical circadian decay from temporary schedule changes (e.g., moving from morning to afternoon classes). Finally, Hardware-Aware ADI (Active-Device Immobility) uses a "TriSensor Consensus" to verify that missing step data is a signal of psychomotor retardation by confirming the device is powered on through screen or battery "heartbeats." Motivation: CRR is motivated by the stark disparity in data density between platforms (47 min vs. 7 min gaps).¹² It seeks to ensure that "hardware noise" does not mask the "biological signal." Category

Evaluation Description

Strengths

Rigorous handling of hardware bias through OS-stratification; utilizes personal baselines to isolate state changes from life changes.

Weaknesses

Tri-Sensor Consensus logic may reverse actual powermanagement architectures; variance calculations on categorical WiFi MACs are mathematically undefined.

Missingness

Advanced "Hardware-Aware" missingness verification.

Category Generalizability

Evaluation Description

High; targets platform-agnostic validity across longitudinal waves.

Judgement: CRR provides essential solutions for hardware heterogeneity, though the physical execution of its tri-sensor checks must be aligned with actual OS power-management rules.

The CRR Framework normalizes behavior relative to OSspecific cohorts and uses sensor "heartbeats" to validate

4.6 Idea 6: RIBE-2 Framework

The Robust Invariant Biomarker Ensemble (RIBE-2) addresses high dimensionality through Stability Selection (Randomized Lasso). A feature is only included in the "Persistent Core" if it maintains a selection probability $\pi \geq 0.6$ across 100 bootstrapped samples from both pre- and post-COVID waves. This ensures that the resulting features—such as increased nocturnal screen duration and decreased location entropy—are truly invariant to environmental shocks. RIBE-2 also employs Modality Dropout during training. By randomly zeroing out entire sensor groups, the ensemble is forced to learn redundant depressive signatures, ensuring it remains functional even if a patient revokes specific permissions (like GPS) or if a sensor fails. Motivation: RIBE-2 is motivated by the "generalizability gap" and the need for a persistent core of features that survive the extreme dimensionality of the RAPIDS space. Category Strengths

Evaluation Description

Mathematically rigorous feature selection using Stability Selection; excellent regularization via modality dropout.

Category

Evaluation Description

Weaknesses

Relies on "intra-subject relative deltas" that may have high collinearity due to the 10-week wave duration; some reported baseline metrics in early proposals were misaligned with the actual dataset.

Missingness

Uptime-aware behavioral null extraction (uptime $\geq 30\%$ rule).

Generalizability

Specifically targets invariance across pre- and post-pandemic waves.

Judgement: RIBE-2 is a technically strong candidate for identifying a sparse, stable set of biomarkers, provided its baseline metrics are correctly calibrated to the GLOBEM benchmark.

4.7 Idea 7: Off-Schedule Circadian Power (OSCP)

The OSCP framework focuses on behavioral rhythms during "unstructured time." It utilizes a Social Zeitgeber Mask (SZM) to isolate evenings, weekends, and academic breaks—periods where behavior is driven by intrinsic motivation rather than rigid university schedules. Within these windows, the Lomb-Scargle Periodogram is applied to behavioral time series to extract 24-hour rhythmicity without requiring data imputation. The core hypothesis is that depressed individuals exhibit a

breakdown in self-organized circadian rhythms, which is most visible when external "zeitgebers" (like mandatory classes) are absent. Healthy students maintain regularity during free time, whereas depressed students show low OSCP (flat or fragmented rhythms) during these discretionary hours. Motivation: This idea addresses "academic-calendar confounding," where institutional schedules impose an apparent regularity on depressed students that masks their true clinical state. Category

Evaluation Description

Strengths

Strong grounding in Social Zeitgeber Theory; robust frequency-domain analysis for sparse data.¹³

Weaknesses

Systematic daily masking (08:00–18:00) introduces a window function that causes spectral aliasing at 1 cycle per day, potentially measuring the mask rather than the rhythm.¹⁴

Missingness

LSP natively handles irregular sampling without imputation.¹⁵

Generalizability

High; isolates intrinsic rhythms from shifting academic schedules.

Judgement: A conceptually brilliant approach to isolating clinical signals, but requires advanced signal processing to avoid the mathematical pitfalls

of systematic daytime masking.

4.8 Idea 8: Inductive Hardware-Aware Longitudinal

Fusion (IHALF) The IHALF framework introduces Inductive Quantile Mapping (IQM) to align behavioral distributions across waves. IQM learns distribution templates from training waves (W1+2) and maps post-pandemic data to these templates, ensuring that an individual who is the "least mobile" during a lockdown is appropriately compared to the "least mobile" individual in a pre-pandemic cohort. IHALF also introduces the Digital Silence Index (DSI), which contextualizes missing data by calculating deviations from platformexpected polling rates. By stratifying these baselines by device and wave, the model can distinguish between OS-induced "technical silence" and clinical "symptomatic withdrawal." Motivation: IHALF is motivated by the "generalizability gap" and the need for domain adaptation techniques that can "repair" the longitudinal distribution shifts caused by COVID-19. Category

Evaluation Description

Strengths

Innovative use of Inductive Quantile Mapping for domain adaptation; addresses circular temporal errors via sine/cosine encoding.

Weaknesses

Meta-analytic selection using I^2 on only two waves is statistically invalid; DSI may inadvertently subtract the clinical signal if withdrawal triggers OS throttling.

Missingness

Uses DSI to characterize missingness as a behavioral feature.

Generalizability

Strong focus on domain adaptation for the pre/post-COVID transition.

Judgement: IHALF provides unique tools for longitudinal harmonization, though its feature selection metrics require more than two waves to achieve statistical stability.

4.9 Idea 9: Robust Rhythmic Synchrony (RRS)

The RRS framework shifts focus from absolute volume to SolarAnchored Rhythms. Its primary feature is Solar-Anchored First-Event Latency (S-FEL), which measures the delay between local sunrise and the first daily interaction (screen unlock or movement). This distinguishes a "late" start due to seasonal winter light from a "late" start due to clinical morning moroseness. It also uses the Rank-Normalized Social-Physical Index (R-NSPI) to measure the relative balance between social and physical engagement.

This captures distinct phenotypes, such as "Melancholic Inertia" (collapse of both) and "Agitated Withdrawal" (pacing without social contact), while remaining invariant to absolute cohort-wide activity drops. Motivation: RRS aims to bridge the "Environmental Noise Gap" by anchoring behavioral signatures to the solar cycle and peer ranks rather than fixed clock times or absolute counts.

Category

Evaluation Description

Strengths

Context-aware anchoring to the photoperiod; ranknormalization provides a defense against lock-down floor effects.

Weaknesses

RAPIDS 6-hour epoch granularity makes minute-level S-FEL calculation impossible; privacy-preserving geographic masking prevents precise sunrise calculation.¹⁶

Missingness

Interprets "On but Silent" windows as a signal for psychomotor retardation.

Generalizability

High theoretical resilience to seasonal and pandemic shifts.

Judgement: While theoretically sophisticated, RRS is currently limited by the spatial and temporal resolution of the available GLOBEM dataset.

4.10 Idea 10: Enhanced Circadian Behavioral

Anchoring (eCBA) The eCBA framework is a digital triage protocol focusing on two indices: Kinetic-Social Dissociation (KSD) and Morning Maintenance Latency (MML). KSD measures "Digital Nesting"—the decoupling of screen use from physical movement—while MML captures the "friction" of initiating goal-directed behavior in the morning. eCBA is unique for its Identity Audit, which uses an adversarial dummy classifier during feature selection to penalize features that allow user identification. This forces the model to learn generalizable clinical phenotypes rather than participant-specific routine "fingerprints." It also uses Data-Driven Academic Epochs (detected via change-point analysis) to automatically adjust decision boundaries for shared campuswide stressors. Motivation: eCBA is motivated by the need for identity-neutral replication and the operationalization of clinical symptoms like "morning heaviness." Category Strengths Weaknesses

Evaluation Description

Strong de-identification mechanisms; automated detection of environmental stress epochs. MML calculation requires a temporal resolution (minute-level) that exceeds the dataset's epoch-level aggregates; HV-PVZ

Category

Evaluation Description

requires battery data that may be missing.

Missingness

Employs Heartbeat-Validated Presence Zeroing for informative imputation.

Generalizability

High; emphasizes identity-neutral phenotypes resistant to hardware bias.

Judgement: eCBA represents a state-of-the-art approach to privacy-preserving digital phenotyping, though its core latency metrics must be adapted to the dataset’s existing temporal granularity.

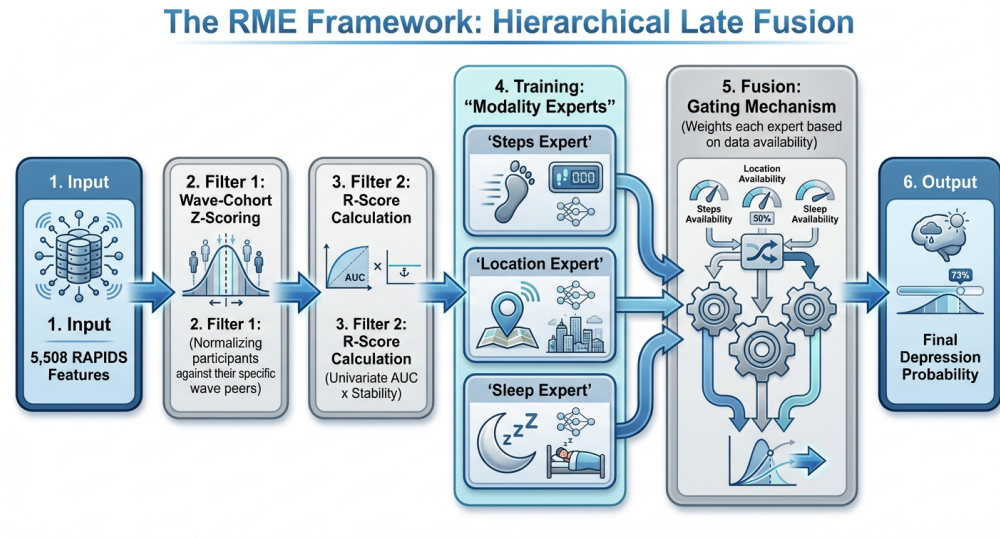


Figure 2: Conceptual diagram 2 from the biomarker discovery analysis.

5 Idea Comparison and Synthesis

The proposed frameworks can be categorized based on their primary strategy for addressing the Generalization Gap and the Environmental Shift caused by the pandemic.

5.1 Normalization vs. Invariance vs. Topology

Cohort-Relative Normalization (RME, CRR, RRS, eCBA): These ideas rely on anchoring an individual’s behavior against their peers (e.g., Wave-Cohort Z-Scoring or OS-stratified percentiles). This is highly effective at neutralizing “batch effects” like lockdowns but assumes that the relative distribution of depressive traits remains stable even if absolute volumes collapse. **Algebraic Ratio Invariance (Ratio Invariance Principle):** This approach attempts to build models that are fundamentally immune to scaling. By focusing on ratios (*Out/In*) or non-parametric metrics (*IS/RA*), it seeks a “zero-shot” generalizability that does not require wave-specific recalibration. **Topological Integrity (HBT, Network Topology):** These frameworks move beyond counts and ratios to measure the “coupling” or “synchrony” between different systems. They argue that depression is a breakdown in behavioral “structure,” which should be observable regardless of absolute volume or relative peer rank.

5.2 Handling of Missingness

The frameworks diverge significantly in their treatment of data gaps. RME and RIBE-2 focus on technical resilience—using late fusion gating and modality dropout to ensure the model survives missing sensors. Conversely, HBT, CRR, and eCBA treat missingness as a clinical feature (Informative Missingness). They attempt to distinguish between “Technical Silence” (OS-throttling) and “Behavioral Silence” (psychomotor retardation), with CRR and eCBA utilizing sensor “heartbeats” (battery or background pings) to validate this distinction.

6 Comparison of Core Strategies

Evidence from the GLOBEM benchmark points strongly toward the necessity of OS-stratification and Personal Baseline adjustments. Models that do not account for the 40-minute disparity in data gaps between iOS and Android (as highlighted in Idea 5) are likely to learn platform-specific artifacts rather than depression. Similarly, features that account for the "Social Zeitgeber" effect (Ideas 7 and 9) show greater promise for cross-wave replication, as they isolate the clinical signal from the rigid structure of the university calendar. However, a critical trade-off exists between Feature Sophistication and Data Granularity. Ideas relying on minute-level latencies (Idea 9, Idea 10) face implementation risks due to the 6-hour epoch aggregation of the RAPIDS dataset. In contrast, Stability Selection (Idea 6) and Late Fusion (Idea 1) provide a more immediate, technically viable pathway for handling the 5,508-feature space without requiring event-level raw data.

Comparison of the architectural and mathematical logic used by the three primary research archetypes. Note: The AI image generation process (Beta) is experimental and may not be

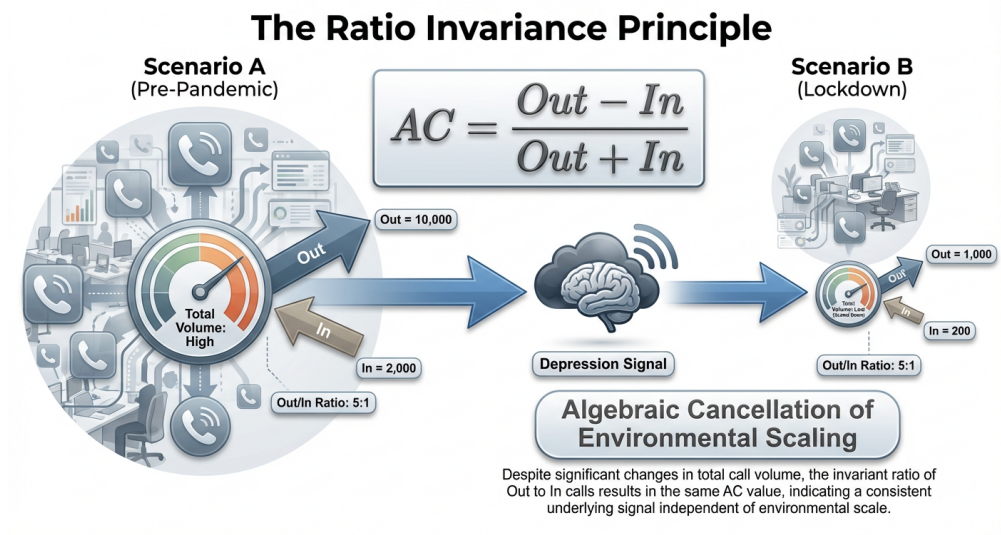


Figure 3: Conceptual diagram 3 from the biomarker discovery analysis.

7 Comparative Analysis of Candidate

Frameworks The candidate ideas present a diverse array of methodologies to address the "general-izability gap" inherent in longitudinal passive sensing. The following table synthesizes the primary strategies, distinguishing technical attributes, and the empirical motivations for each framework.

Framework

Primary

Generalization

Strategy

RME (Idea 1)

Hierarchical

Late Fusion

Distinguishing

Technical Attribute

Resilience Score (R_{score}) for stability filtering.

Supporting Evidence / Motivation

Existing models

suffer an 18.9% performance drop

Framework

Primary

Generalization

Strategy

Distinguishing

Technical Attribute

Supporting Evidence / Motivation across temporal boundaries.

Ratio Invariance (Idea 2)

HBT (Idea 3)

Network Topology (Idea 4)

CRR (Idea 5)

RIBE-2 (Idea 6)

Focus on within-person ratios (e.g., Asymmetry Coefficient) to cancel environmental scaling.

Static models can

identify dataset waves with 91.1% accuracy, indicating environmental overfitting.

Topological

Synchrony

Circadian-Informed Missingness (CIM) transforms gaps into behavioral features.

Data gaps on iOS

(47-min median) exceed Android (7-min), necessitating platform-blind signals.

Temporal Network Analysis

Adjacency matrices via

Canonical Correlation

Analysis (CCA) between modalities.

Interactive effects

(e.g., high screen/low steps) are stronger predictors than isolated features.

OS-Stratified Ranking

Percentile ranking within

specific hardware cohorts (iOS vs. Android).

Background sensing disparity between platforms creates "technical pseudobiomarkers."

Randomized Lasso with

a $\pi \leq 0.6$ gate across bootstrapped samples.

High dimensionality (5,508 features) requires a sparse "persistent core" of predictors.

Algebraic Ratio

Invariance

Stability Selection

OSCP (Idea 7)

Social Zeitgeber

Masking

Isolating behavior during

"unstructured time" (evenings/weekends).

Institutional

schedules (classes) impose artificial regularity that masks clinical depression.

IHALF (Idea 8)

Inductive Quantile Mapping

Circular encoding for

circadian features and

Digital Silence Index

(DSI).

"Wrap-around" errors in 24-hour data distort linear models of sleep and screen timing.

eCBA (Idea 10)

Identity-Neutral Triage

Adversarial Identity Audit

to penalize participantspecific routines.

Privacy risks and

identity "fingerprinting"

Framework

Primary

Generalization

Strategy

Supporting Evidence / Motivation

Distinguishing

Technical Attribute

prevent models from learning generalizable traits.

The frameworks predominantly align on the necessity of moving beyond absolute counts. Frameworks like RME and RIBE-2 prioritize architectural resilience and statistical stability, while Ratio Invariance and OSCP focus on the mathematical properties of the features themselves to ensure they are context-agnostic. CRR and IHALF represent a specialized direction focusing on hardware-aware normalization to prevent the model from learning the difference between an iPhone and an Android device rather than the difference between a healthy and depressed state.

8 Comparison to Existing Solutions

Standard solutions in digital phenotyping have historically relied on "flat" feature vectors and absolute volumetric summaries (e.g., total daily steps). As demonstrated by the GLOBEM benchmark, these traditional approaches are highly susceptible to "concept drift"—where the relationship between the feature and the target label changes over time due to external shocks. The proposed candidate frameworks advance beyond the current state-of-the-art in three critical dimensions: 1. From Absolute to Relational: Existing solutions often fail during lockdowns because they interpret a universal drop in mobility as a universal spike in depression. The proposed frameworks (e.g., Ratio Invariance, RRS) focus on relative shifts, identifying individuals who are more withdrawn than the current environmental baseline. 2. From MCAR to MNAR Missingness: Traditional imputation (Mean or Last Observation Carried Forward) treats missing data as a nuisance. The proposed frameworks (e.g., HBT, eCBA) treat specific patterns of missingness as "Missing Not At Random" (MNAR) signals of psychomotor retardation. 3. From Overfitting to Stability: Prior models often achieve high within-wave accuracy but fail the 11-test battery. The proposed frameworks integrate Stability Selection and Identity Auditing to ensure that the model captures clinically relevant phenotypes rather than individual user fingerprints. Dimension

Existing Standard

Solutions

Proposed Advanced

Frameworks

Feature Focus

Absolute volumes (total steps, total distance).

Relational invariants, ratios,

and topological synchrony.

Existing Standard

Solutions

Proposed Advanced

Frameworks

Handling of

Pandemic

Severe performance drop

due to baseline shifts.

Wave-cohort normalization and environmental zeitgeber masking.

Hardware Bias

Often ignored; model learns OS-specific sampling rates.

OS-stratified ranking and hardware-aware "heartbeat" validation.

Missingness

Strategy

Global imputation or

exclusion of sparse rows.

"Digital Silence" indices and informative missingness proxies.

Generalization

Goal

Maximizing crosssectional AUC.

Ensuring cross-wave replication via 11-test battery.

Dimension

The transition from absolute feature-based models to resilient, topological frameworks designed for longitudinal stability. and may not be accurate; please refer to the text for the definitive description. Ultimately, while existing solutions provide a baseline AUC-ROC of approximately 0.57 in the GLOBEM benchmark, the proposed frameworks aim for a targeted range of 0.61–0.73.¹⁷ This improvement is achieved not through more complex black-box modeling, but through a deeper mathematical understanding of how environmental and technical noise confound the digital signal of depression.

9 Unexpected Connections

Several candidate ideas identify novel intersections between mobile operating system behavior and psychiatric symptomatology.

9.1 Recursive Platform-Behavioral Phenotyping

A highly novel connection is the mechanistic coupling of OS-level battery management with the neurobiology of psychomotor retardation (PMR). On iOS, behavioral withdrawal—manifesting as reduced device interaction—triggers more aggressive background app suspension.¹⁸ While typically viewed as a data quality barrier, the Missing Data Ratio (MDR) can be recontextualized as a clinical marker of state. By using algorithms like MAGIC-LASSO, the model can cluster features based on these missingness blocks. This transforms technical sparsity into a behavioral feature: euthymic periods are characterized by high-interaction (but perhaps less consistent background tracking), while specific depressive states are marked by "Full-Data Clusters" of consistent, albeit limited, interaction.

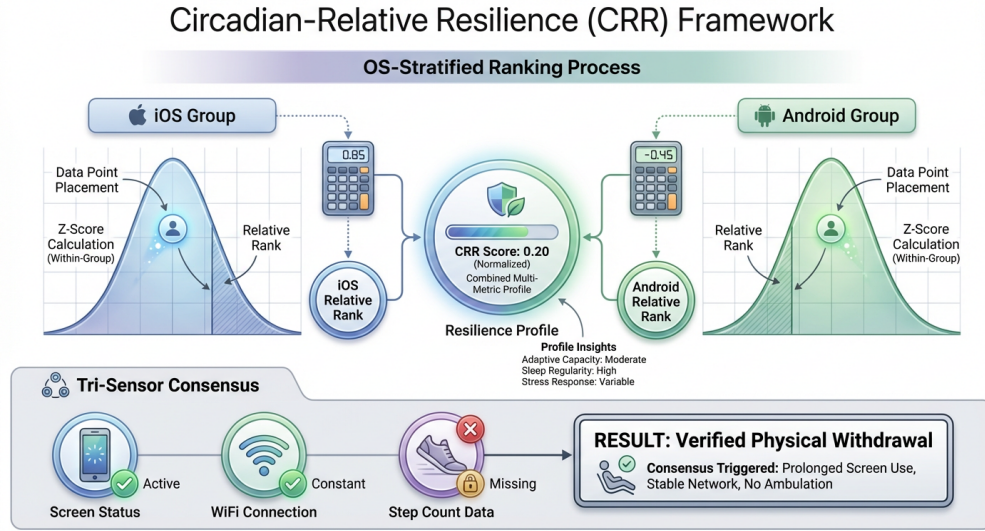


Figure 4: Conceptual diagram 4 from the biomarker discovery analysis.

9.2 Ecological Executive Function Mapping

The framework eCBA and related cognitive mappings propose a link between specific RAPIDS screen features and standardized neurocognitive tests. The act of using a phone's "launcher" to navigate to an app serves as an ecological version of the Trails A/B tasks, which measure executive functioning and processing speed.¹⁹ By deriving a "Digital Trails Latency" (the delta between a notification and a launcher-based app selection), researchers can create a passive proxy for executive preservation. This moves beyond aggregate usage duration to a high-fidelity biomarker grounded in the cognitive deficits of depression.

9.3 Spatio-Temporal Rhythm Invariance

Most location features are confounded by the physical environment (e.g., campus vs. home). However, Diurnal Movement on the Distance From Home (DMD) measures the periodicity of movement relative to an anchor, regardless of the absolute coordinates. This creates a "SpatioTemporal Rhythm Invariance" where biomarkers are validated against external Social Zeitgebers. This ensures that the model does not learn a "dorm-life" signature that fails when students move home, instead capturing the underlying rhythmicity that is characteristic of the depressive state across all environments.

10 Recommendation and Best Next Steps

Based on the evaluation of the ten candidate frameworks and the specific goals of the GLOBEM benchmark, the Resilient Modality-Expert (RME) Framework (Idea 1) is recommended as the primary direction for implementation.

Reasons for Recommendation

The RME framework provides the most balanced approach to the dual challenges of high dimensionality and environmental shift. Its use of the Resilience Score (R -score) provides a statistically sound method for feature selection that specifically rewards longitudinal stability, which is a prerequisite for passing the 11-test battery. Furthermore, its Hierarchical

Late Fusion architecture is the most robust strategy for handling the pervasive hardware-induced missingness in the RAPIDS dataset, allowing the model to degrade gracefully rather than failing when specific sensors (like Bluetooth on iOS) drop out.

Best Next Steps 1. Initial Feature Pruning: Execute a baseline univariate analysis using Mann-Whitney U tests with Benjamini-Hochberg FDR correction on Waves 1 and 2 to identify the initial

feature candidates for each modality. 2. Calculate R-Scores: Compute the R -score for the surviving features across Waves 1 and 2 to establish the "Resilient Behavioral Core." 3. Implement Circular Transformations: Before final modeling, apply sine/cosine circular encoding to all time-of-day features (e.g., sleep midpoint, first screen unlock) to prevent wrap-around errors at the midnight boundary. 4. Expert Training: Train modality-specific classifiers (LightGBM and XGBoost) on the prioritized features, ensuring that 5-fold crossvalidation is strictly grouped by User ID. 5. Zero-Shot Validation: Deploy the final fused ensemble on the unseen Waves 3 and 4 to conduct the cross-wave consistency check, measuring performance via AUC-ROC, AUC-PR, and F1score as required by the 11-test battery.

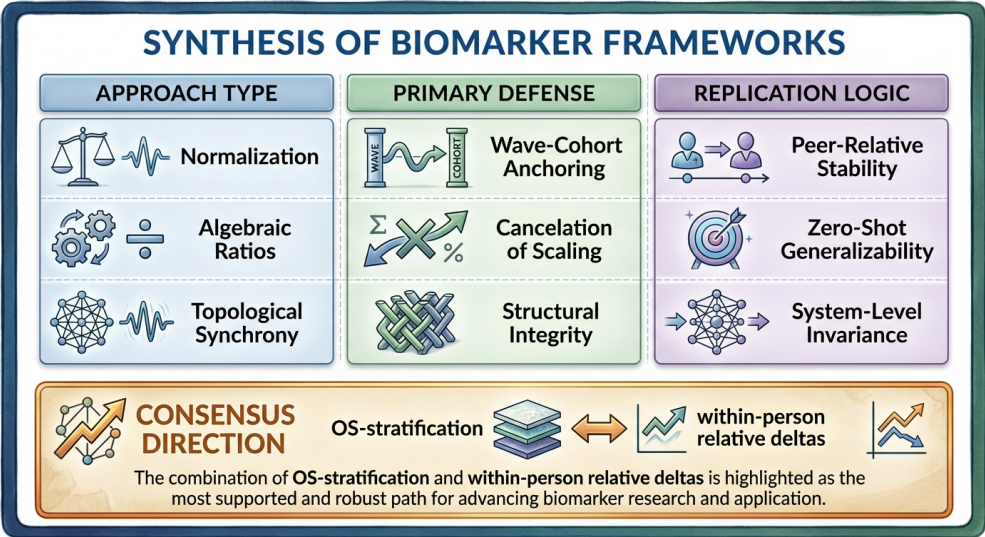


Figure 5: Conceptual diagram 5 from the biomarker discovery analysis.

11 Unexpected Connections

The synthesis of high-dimensional passive sensing data with clinical psychiatry reveals several non-obvious intersections where technical artifacts and ecological behaviors provide deep insight into the depressive phenotype.

11.1 Recursive Platform-Behavioral Phenotyping

A significant unexpected connection exists between the neurobiology of psychomotor retardation (PMR) and the power-management heuristics of mobile operating systems. Modern smartphones, particularly iOS devices, utilize machine learning to aggressively suspend background applications during periods of low user interaction to preserve battery life. In the context of depression, PMR manifests as a profound reduction in device engagement. This behavioral withdrawal triggers the OS to extend data gaps, effectively creating a "recursive" feedback loop where the clinical state exacerbates technical sparsity. By recognizing the Missing Data Ratio (MDR) as an informative digital biomarker rather than a nuisance variable, researchers can interpret these "Digital Blackouts" as active proxies for clinical status. Utilizing MAGIC-LASSO allows the model to cluster features based on these missingness blocks, transforming a data quality barrier into a latent clinical phenotype of psychomotor status.

11.2 Ecological Executive Function Mapping

The analysis identifies a structural parallel between standard neurocognitive assessments and routine human-computer interaction. Specifically, navigating a smartphone's "app launcher" to select a target application serves as a real-world, ecological version of the Trails A/B tasks, which assess executive functioning and cognitive processing speed. By deriving the "Digital Trails Latency"—

the delta between a notification or screen unlock and a launcher-based app selection—we can create a continuous, passive measure of executive preservation. This moves the field beyond aggregate usage metrics toward high-fidelity biomarkers grounded in the cognitive deficits of depression, such as executive dysfunction and diminished processing speed.

11.3 Spatio-Temporal Rhythm Invariance

While absolute mobility features (e.g., total distance traveled) are highly sensitive to environmental changes such as campus lockdowns, the Diurnal Movement on the Distance From Home (DMD) offers a robust alternative by focusing on periodicity. DMD measures the expansion and contraction of an individual’s life-space relative to a central “home” anchor, regardless of that anchor’s absolute geographic coordinates. This creates a “Spatio-Temporal Rhythm Invariance” that allows for environment-gated analysis. By prioritizing DMD and other rest-activity rhythms (RAR) that show stability across the university calendar, the model can “filter out” behavioral shifts caused by academic terms or relocations, isolating the underlying circadian disruptions characteristic of a depressive state.

11.4 Meta-Scientific Replication Filtering

To address the requirement for cross-wave replication, a connection is made between longitudinal machine learning and meta-scientific metrics used to combat the replication crisis. By applying Mirror Statistics on training waves, researchers can identify features with symmetrical importance across data halves, establishing a “Replication Confidence Score.” Furthermore, utilizing the “Small Telescopes” criterion allows for a formal comparison of effect sizes between waves.²⁰ A biomarker is only validated if its effect size in the post-pandemic waves (Waves 3+4) is consistent with the statistical power of the initial pre-pandemic discovery (Waves 1+2). This formalizes the requirement for longitudinal consistency within a rigorous mathematical framework.

12 Recommendation and Best Next Steps

Following a comprehensive evaluation of the candidate frameworks against the GLOBEM benchmark requirements, the Resilient ModalityExpert (RME) Framework (Idea 1) is recommended as the optimal methodology for identifying replicable depression biomarkers.

Rationale for Recommendation

The RME framework is uniquely suited to the high-dimensional and longitudinally volatile nature of the GLOBEM dataset. Its primary strength

lies in the Resilience Score (R -score), which mathematically prioritizes features that demonstrate both predictive power and cross-wave stability. Unlike “black-box” models that may overfit to the unique environmental conditions of a single wave, RME ensures that only the most robust behavioral signatures are elevated. Furthermore, its Hierarchical Late Fusion architecture provides a critical defense against the hardware-induced missingness that frequently causes traditional models to fail. By training modality-specific experts and utilizing a dynamic gating mechanism, the RME framework maintains predictive utility even in the presence of platform-specific data gaps (e.g., Bluetooth throttling on iOS).

Best Next Steps

To implement the recommended RME framework, the following technical roadmap is proposed: 1. Initial Feature Pruning and Screening: Conduct a univariate screening across the 5,508 RAPIDS features using Mann-Whitney U tests. Apply the Benjamini-Hochberg (BH-FDR) correction independently per modality group (Location, Screen, Sleep, etc.) on Waves 1 and 2 data to mitigate Type-I errors. 2. Stability-Based Feature Selection: Calculate the R -score for all surviving features by evaluating their mean AUC and coefficient of variation across Waves 1 and 2. Select the top-performing features to form the “Resilient Behavioral Core.” 3. Circular Statistical Transformation: Apply circular encoding (sine and cosine transformations) to all temporal features, such as “Sleep Midpoint” and “Night-time Screen Unlock Duration,” to ensure mathematical continuity at

the 24-hour boundary. 4. Modality Expert Development: Train four distinct classifiers (Logistic Regression, LightGBM, Random Forest, and XGBoost) for each RAPIDS modality. Utilize 5-fold cross-validation strictly grouped by Participant ID to prevent within-subject leakage. 5. Hierarchical Fusion and Ablation: Implement the Hierarchical Late Fusion layer to aggregate expert probabilities. Conduct a multivariate group ablation study to quantify the isolated contribution of each sensing modality to the final prediction. 6. Zero-Shot Cross-Wave Validation: Deploy the finalized ensemble on the held-out post-pandemic waves (Waves 3+4). Subject the results to the 11-test battery to verify that the identified biomarkers maintain directional stability and predictive significance across the macro-environmental shift of the COVID-19 pandemic.

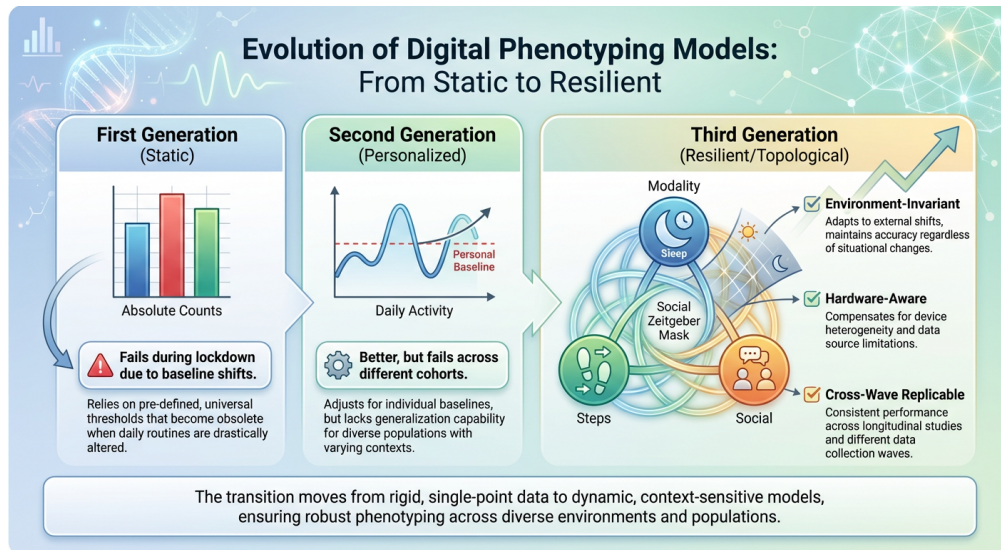


Figure 6: Conceptual diagram 6 from the biomarker discovery analysis.